

# Brownian free motion: the *velocitas ultima* denied, one action constant, and the floor

**After refereeing (GPT-6 Astra, 2026-09-28).** A7: ACCEPT S1's continuous-Lévy classification and nowhere differentiability, S2's nonnegative additive-function proof, and S3's Cameron–Martin norm and optimal-test constant. REFINED the hypotheses: deterministic initial position, stationary independent increments, isotropy of centred noise, all positive masses, and additive force-independent position noise in S3. REJECT the unqualified claim that Nelson proves independence from the literal Newtonian Laws or equivalence under P1–P5 alone. His mean dynamics and its additional state and phase assumptions are now separated from this model. The §5 table and scope are correspondingly qualified;  $\kappa > 0$  remains assumed. The bounded prior-art search in §3 located the centre-of-mass variance calculation in Demme–Caticha, with the inverse-mass law supplied there. The ancillary jump discussion now names stochastic continuity and qualifies its dimensional argument.

**Result, 2026-09-28 (Claude; refereed with corrections above).** A third route for the fifth-postulate programme, beside the deformation route (Theorems A, B) and the Gaussian state route (Theorem B'): the stochastic one. Let the free motion of a body of mass  $m > 0$  start at a deterministic point and have continuous paths, stationary independent increments and isotropic centred noise (premises P1–P3 of §1).

- **Theorem S1 (the law and the denial).** The free motion is Brownian motion with drift, with variance rate  $D(m) \geq 0$  per coordinate. For  $D = 0$  it is Newton's inertial motion. For  $D > 0$  the paths are almost surely nowhere differentiable, so the *velocitas ultima* that Newton's scholium calls “certus & definitus” exists only as a mean velocity.
- **Theorem S2 (one constant).** If the law depends on the mass alone (P4) and the centre of mass of independent bodies obeys the law of their total mass (P5), then  $\kappa = mD$  is one universal constant, with the dimension of an action;  $\kappa = 0$  is admissible.
- **Theorem S3 (the floor, additive-noise hypothesis of §4).** For  $\kappa > 0$ , any record that decides from the path, with pinned ends, between the inertial chord and the constant-force parabola over a cell errs at equal priors with probability at least  $\Phi(-\sqrt{K_\tau/2\kappa})$ , attained by the optimal test, where  $K_\tau = F^2\tau^3/(24m)$  is the Galileo action of the cell. Confidence  $1 - \epsilon$  needs  $K_\tau \geq 2z_{1-\epsilon}^2\kappa$ ; with  $\kappa = \hbar$  this is the mark mesh of the Planck paper, eq. (2).
- **Proposition S4 (jumps allowed: the leap).** Dropping continuity, P2–P5 force the Lévy exponent of a body of mass  $m$  to be  $m\Gamma(|k|/m)$  for one universal function  $\Gamma$ . If the only universal constant is an action  $\kappa$ , dimensional analysis forces  $\Gamma(p) = C\kappa p^2$ , and continuity follows. With a speed  $c$  as well, the relativistic kinetic energy divided by  $\kappa$ ,  $(\sqrt{m^2c^4 + c^2\kappa^2k^2} - mc^2)/\kappa$ , is of this form and generates a process with jumps.
- **Mean-law compatibility (Nelson).** With diffusion coefficient  $\nu = \kappa/2m$ , Nelson's stochastic mechanics imposes Newton's second law on a symmetric mean acceleration and obtains Schrödinger dynamics with additional state and phase assumptions and  $\hbar = \kappa$  (Nelson 1966, abstract; precursor Fényes 1952, metadata). This supports a separate mean-law interpretation for  $\kappa > 0$ ; the deterministic  $\kappa = 0$  branch supplies ordinary free motion.

The route addresses the fifth-postulate goal through the statement (the existence of the ultimate velocity), compatibility with specified mean laws, one added action constant, and a floor on recording the inertial–parabola difference, an area Newton takes to zero. Its limit is the same as in the other routes:  $\kappa > 0$  is the negated postulate, assumed, and the zero branch stays admissible. Theorems S1 and S2 are elementary; §3 records prior art without a novelty claim.

## 1. Premises

- **P1 (continuity).** The free motion starts at a deterministic position and has almost surely continuous paths. Continuity of motion is Aristotle's (*Physics* VI, cited by book only);

the leap of al-Nazzam, already in the Planck paper’s classics, would be a jump, which P1 excludes.

- **P2 (increments).** Stationary independent increments. This explicitly includes the independence premise beyond space and time homogeneity. A Galilean boost adds a constant drift and preserves P2.
- **P3 (isotropy).** The centred fluctuations have a rotation-invariant law; the specified constant drift may select a direction. Isotropy of the full law in a fixed frame would force that drift to vanish.
- **P4 (mass alone).** The centred free law depends only on mass and is defined for every real  $m > 0$ , with finite variance rate in S1–S2. Universality of free fall alone does not establish this noise premise.
- **P5 (composition).** For independent bodies of any positive masses, their centre of mass has the centred free law of the total mass and the mass-weighted mean drift. Correlated noises require a new premise.

## 2. Theorem S1 (ACCEPT with the stated P1–P3)

A process in  $\mathbb{R}^3$  with stationary independent increments and continuous paths is a Brownian motion with drift,  $X_t = x_0 + vt + \Sigma^{1/2}W_t$  (the Lévy–Itô decomposition has no jump part when the paths are continuous; Lalley, *Introduction to Brownian Motion*, §1, passage). Continuity supplies stochastic continuity, and stationary independent increments supply the Markov property. Isotropy of the centred law gives  $\Sigma = DI$ , and  $D \geq 0$  can depend only on the body. For  $D > 0$ , Brownian paths are almost surely nowhere differentiable (Paley, Wiener and Zygmund 1933, metadata), and their spatial range in  $\mathbb{R}^3$  over any nontrivial time interval has Hausdorff dimension 2, the dimension compared for quantum paths by Abbott and Wise (1981) (metadata). The difference quotient  $(X_{t+h} - X_t)/h$  has no limit; its conditional mean does,  $\lim_{h \downarrow 0} E[X_{t+h} - X_t | \mathcal{F}_t]/h = v$ , which is Nelson’s mean forward velocity.  $\square$

In the words of the scholium closing Book I, Section I (quoted from the 1687 text in §1 of the fifth-postulate note): “Extat limes quem velocitas in fine motus attingere potest, non autem transgredi. Hæc est velocitas ultima. [...] Cumq; hic limes sit certus & definitus”. For  $D > 0$  the limit of the ratio exists for the mean and fails for the path. This is the denial of joint determinacy in the form Newton wrote it: the place at an instant is sharp, while the velocity at that instant has no pathwise value.

## 3. Theorem S2 (ACCEPT under P4–P5)

*Proof.* Take independent bodies with masses  $m_i$ , drifts  $v_i$  and variance rates  $D(m_i)$ , and total mass  $M = \sum m_i$ . The centre of mass  $X = \sum m_i x_i / M$  has the constant drift  $\sum m_i v_i / M$  (momentum conservation) and the variance rate  $\sum m_i^2 D(m_i) / M^2$ . By P5 this equals  $D(M)$ , so the function  $g(m) = m^2 D(m)$  satisfies  $g(m_1 + m_2) = g(m_1) + g(m_2)$  for all  $m_1, m_2 > 0$ . Since  $g \geq 0$ ,  $g$  is nondecreasing, and Cauchy’s equation gives  $g(m) = \kappa m$ . Hence  $D(m) = \kappa / m$  and  $mD = \kappa$  is the same for every body. Its dimension is mass times length<sup>2</sup>/time, an action.  $\square$

For completeness, nonnegativity gives  $g(y) - g(x) = g(y - x) \geq 0$  for  $y > x > 0$ . Fix  $m_0 > 0$ . Additivity gives  $g(r m_0) = r g(m_0)$  for positive rational  $r$ ; monotone rational approximations to  $m/m_0$  give  $g(m) = m g(m_0)/m_0$ . Thus no measurability assumption or empirical mass-continuity assumption is needed once P4–P5 cover all positive real masses. A restricted list of species or correlated noises would need a separate argument.

The argument is the stochastic counterpart of the rotation-composition note, where interaction forces a common constant in the composition of rotations; P4 is an additional universality premise for fluctuations, and P5 specifies the centre-of-mass law. Neither argument excludes  $\kappa = 0$ .

**Prior art (bounded search, 2026-09-28).** Nelson 1966, abstract, explicitly supplies  $\nu = \hbar/(2m)$ . Demme–Caticha, *The Classical Limit of Entropic Quantum Dynamics* (2017), passage, §3, eqs. (18)–(28), computes the centre-of-mass covariance  $\langle \Delta W^a \Delta W^b \rangle = \eta \Delta t \delta^{ab}/M$  from constituent covariances  $\eta \Delta t/m_i$  given in its eq. (9). That is close prior art for the composition calculation; S2 reverses the implication under P4–P5 by the nonnegative Cauchy equation. Searches for stochastic mechanics, inverse mass and centre-of-mass consistency located no explicit version of that converse in the passages checked. This bounded search supports no priority claim. The Demme–Caticha DOI and title were verified through Crossref.

#### 4. Theorem S3 (ACCEPT for additive noise)

In addition to S1–S2 assume explicitly that, under constant force  $F$ , the body’s path is the deterministic Newtonian parabola plus the same additive noise  $\sqrt{D} W_t$ . Take  $m, \tau, \kappa > 0$ , equal priors and  $0 < \epsilon < 1/2$ . Pin both laws to the same endpoints of a cell of duration  $\tau$ . The two laws to be told apart, with and without the force, are Brownian bridges of variance rate  $D = \kappa/m$  around the parabola and around the chord. Their difference is the shift  $\delta x$  (chord minus parabola), with Cameron–Martin norm  $\|\delta x\|^2 = D^{-1} \int_0^\tau \dot{\delta x}^2 dt = (m/\kappa) \int \dot{\delta x}^2 = 2K_\tau/\kappa$ . The Neyman–Pearson test with equal priors errs with probability  $\Phi(-\frac{1}{2}\|\delta x\|) = \Phi(-\sqrt{K_\tau/2\kappa})$ , and no test does better. This is Proposition 6 of the cut-measure note with  $\hbar$  replaced by  $\kappa$ ; its Theorem 2 distributes the evidence  $K_\tau/\kappa$  over any sequence of cuts, each cut carrying the Kullback–Leibler share  $3s(1-s)K_L/\kappa$ .  $\square$

The pinned covariance is  $D(\min(u, v) - uv/\tau)$  and the shift is  $\delta x(u) = Fu(\tau - u)/(2m)$ . Hence  $\int_0^\tau \dot{\delta x}^2 du = F^2 \tau^3/(12m^2)$ , fixing the factor 2 in the norm. Any record obtained from the path by a common observation channel has no smaller Bayes error than the full-path test. For  $F = 0$  that error is  $1/2$ ; the displayed mesh assumes  $F \neq 0$ .

With  $\kappa = \hbar$ , which is the normalization of the Euclidean free measure  $e^{-S/\hbar}$  and of Nelson’s  $\nu = \hbar/2m$ , the threshold is  $\tau \geq (48z_{1-\epsilon}^2 m \hbar / F^2)^{1/3}$ , the Planck paper’s eq. (2).

#### 5. Where the route stands

| part of the goal                      | deformation route              | state route           | stochastic route                                |
|---------------------------------------|--------------------------------|-----------------------|-------------------------------------------------|
| statement changed                     | commutativity (Thm A)          | sharp states (Thm B') | <i>velocitas ultima</i> exists pathwise (S1)    |
| compatibility with specified dynamics | Moyal (Thm A)                  | Gaussian closure      | mean laws; Nelson requires further assumptions  |
| exactly one constant                  | Gutt, under covariance (Thm B) | Thm B'                | composition, P4–P5 (S2)                         |
| floor on the comparison               | Thms C, E                      | Thm C                 | Cameron–Martin for additive position noise (S3) |

**Nelson claim: REFINE; unqualified independence/equivalence: REJECT.** The 1966 abstract reports a stochastic Newton law and an equivalence within a limited framework. Nelson uses forward and backward mean derivatives and a state-dependent drift; P1–P5 alone specify neither that dynamics nor a wave function. Global equivalence also requires phase/circulation conditions (Wallstrom 1994, abstract; DOI verified through Crossref). Thus the literal pathwise Laws, which use an instantaneous momentum, require a reformulation on Brownian paths. In the additive model of S3 one does have  $m d^2 E[X_t]/dt^2 = F$  before endpoint conditioning. That elementary mean identity and Nelson’s symmetric mean acceleration are distinct compatibility statements. The table records their qualified role.

The selection question in all three columns is why the constant is positive. In the stochastic route it has the plainest form, whether free paths have an instantaneous velocity, and Newton's scholium asserts that they do. Two further limits: P4 and P5 are premises about bodies (an equivalence principle for fluctuations and a centre-of-mass law), and S3 is proved for additive position noise, the law of S1 with a force added; in Nelson's full dynamics the drift depends on the state, and the corresponding bound is not claimed here. Allowing jumps (the leap) is treated in §5b.

## 5b. Jumps allowed (Proposition S4)

Replace P1 by stochastic continuity and a càdlàg version, retain the deterministic initial point, and keep P2–P5 without a finite-variance requirement. The free law of a body of mass  $m$  is then an isotropic Lévy process; take it symmetric, so its exponent,  $E e^{ik \cdot (X_t - X_0)} = e^{-t \phi(m, |k|)}$ , is real and nonnegative. The centre of mass of independent bodies has exponent  $\sum_i \phi(m_i, m_i |k|/M)$ , and P5 requires  $\phi(M, |k|) = \sum_i \phi(m_i, m_i |k|/M)$ . Put  $|k| = Mp$  and  $G(m, p) = \phi(m, mp)$ : then  $G(m_1 + m_2, p) = G(m_1, p) + G(m_2, p)$ , and  $G \geq 0$  gives  $G(m, p) = m \Gamma(p)$  with  $\Gamma(p) = G(1, p)$ . Hence

$$\phi(m, |k|) = m \Gamma(|k|/m), \quad \nu_m(A) = m \nu(mA),$$

where  $\nu$  is the Lévy measure of  $\Gamma$ : a body of mass  $m$  jumps with the universal jumps shrunk by  $1/m$  and their rate multiplied by  $m$ . Brownian motion is  $\Gamma(p) = \kappa p^2/2$ , giving  $D = \kappa/m$  as in S2, and the symmetric  $\alpha$ -stable laws are  $\Gamma(p) = \kappa_\alpha p^\alpha$ , with scale  $\kappa_\alpha m^{1-\alpha}$ .  $\square$

*Dimensional analysis.*  $\phi$  is an inverse time and  $p = |k|/m$  has dimension  $(\text{mass} \cdot \text{length})^{-1}$ , so  $\Gamma$  has dimension  $(\text{mass} \cdot \text{time})^{-1}$ . If the only universal constant is one action  $\kappa$ , the unique monomial of that dimension is  $\kappa p^2$ , and no dimensionless combination of  $\kappa$  and  $p$  exists; so  $\Gamma(p) = C \kappa p^2$  and the law is Brownian. Continuity is then a consequence: a jump needs a second constant. With a speed  $c$  available,  $\kappa p/c$  is dimensionless and  $\Gamma(p) = \kappa p^2 f(\kappa p/c)$  is allowed. The choice  $\Gamma(p) = (\sqrt{c^4 + c^2 \kappa^2 p^2} - c^2)/\kappa$  gives  $\phi = m \Gamma(|k|/m) = (\sqrt{m^2 c^4 + c^2 \kappa^2 k^2} - mc^2)/\kappa$ , the relativistic kinetic energy divided by  $\kappa$ , which reduces to  $\kappa k^2/2m$  for  $\kappa |k| \ll mc$ ; it is a Bernstein function of  $k^2$ , and its process has jumps (Carmona, Masters and Simon 1990, metadata). Here  $c$  is an additional speed scale in a Euclidean generator; its jump process allows instantaneous spatial jumps and supplies no causal speed bound on sample paths. The single-action dimensional conclusion is conditional on the absence of every other dimensional input or hidden scale and on dimensional covariance of the law. Those premises, including Aristotle's proposed continuity comparison, are structural interpretations rather than consequences of composition alone.

## 6. Consequence for STATE

Newton necessity gains a third route in which the negated postulate is Newton's own sentence on the ultimate velocity: continuous stationary independent increments with isotropic centred noise give Brownian free motion, composition over all positive masses gives one action constant, and the Cameron–Martin theorem gives the floor, equal to the Planck paper's mark mesh at  $\kappa = \hbar$ , under additive-noise hypotheses. S1–S3 are refereed with these qualifications. Positivity of  $\kappa$  and a physical justification of the path and record laws remain premises.